

Introduction to changepoint detection

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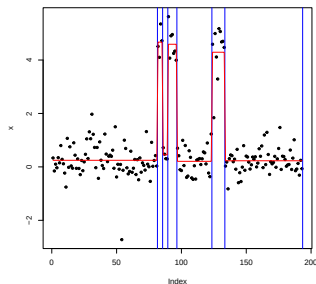
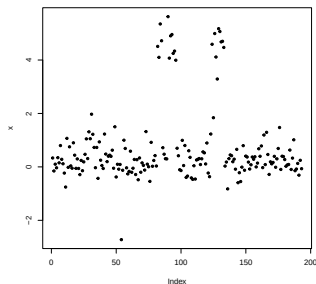
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Introduction

- ▶ Detecting and locating changes in distribution within time series data presents a fundamental statistical challenge.
- ▶ The first studies on changepoint detection emerged in the 1940s [Wald, 1945, Page, 1954]
- ▶ A significant increase in research activity in this area has occurred in recent decades (see [Venkatraman and Olshen, 2007, Killick et al., 2012, Fryzlewicz, 2014, Maidstone et al., 2017] among many others))

A Simple Example



- ▶ Data from [Lai et al., 2005]
- ▶ Comparative Genomic Hybridization (CGH) reveals chromosomal aberrations in DNA.
- ▶ Abrupt changes in signal intensity reveal these aberrations.

Statistical Goals

Detection Has a change occurred?

How many If there are changes, how many are there?

No Spurious Avoid the detection of false changes.

Estimation If there is a change, what is the intensity before and after?

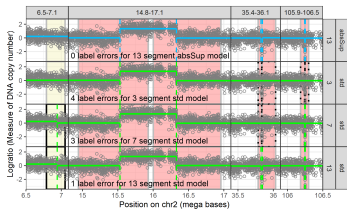
Localization If there are changes, where are they, and how confident are we about their locations?

Many Types of Changes

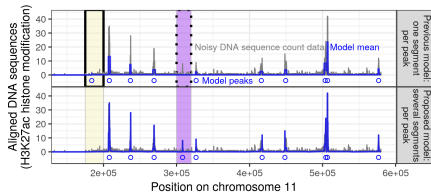
- ▶ Gaussian data: Change in mean.
- ▶ Genomics: Poisson or Negative Binomial models
- ▶ Slope
- ▶ Peaks
- ▶ Changes in variance
- ▶ Changes in multiple parameters.
- ▶ Network structure changes.
- ▶ Non-parametric changes in distribution
- ▶ ...

Many Types of Changes

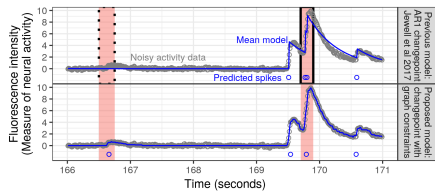
CGH



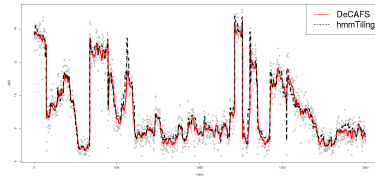
Chip-Seq



Neuroscience



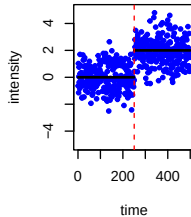
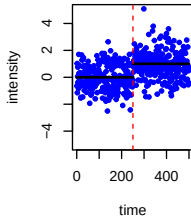
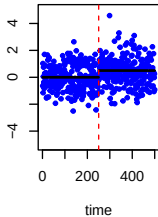
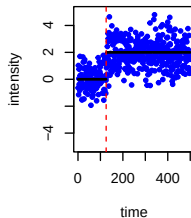
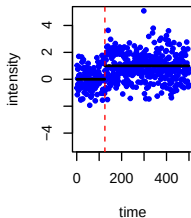
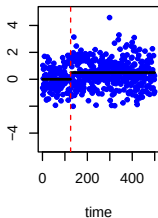
RNA-seq



The univariate change-in-mean model is not that simple!

- ▶ Despite its apparent simplicity, it remains an active area of research (see, for example, [Killick et al., 2012, Aue and Kirch, 2024, Fryzlewicz, 2014, Kovács et al., 2023, Verzelen et al., 2023, Yu et al., 2023])
- ▶ Let us consider the “vanilla” approach for just **one change**.
 - ▶ Our data $y_1, y_2, \dots, y_n$
 - ▶ Assuming the $y_t = \mu_t + \varepsilon_t$
 - ▶ Assuming ε_t are Gaussian $\mathcal{N}(0, \sigma^2)$
 - ▶ $\mu_t = 0$ for $t \leq \tau^*$ and δ otherwise
- ▶ Objectives: Has a change occurred ? Where ?

At most one changepoint illustration



Stop and Think

Stop and Think

- ▶ How many possible segmentations are there?
- ▶ How does the difficulty of detecting a changepoint depend on the height of the change and the position of the changepoint?
- ▶ How can we proceed to find the changepoint, if it exists?

At most one change “vanilla” approach

- ▶ Compare the mean square error with a change at τ :

$$\sum_{t=1}^{\tau} (y_t - \bar{y}_{1:\tau})^2 + \sum_{t=\tau+1}^n (y_t - \bar{y}_{\tau+1:n})^2,$$

- ▶ with the mean square error without a change

$$\sum_{t=1}^n (y_t - \bar{y}_{1:n})^2$$

- ▶ A large difference between these two indicates a change
- ▶ *Maximizing the likelihood ratio test statistic* ($LR_{\max}$)

Why is it not simple?

- ▶ We need to consider $n - 1$ changepoints/models
- ▶ To decide/infer whether there is a change or not
 - ▶ Compute/Compare all $n - 1$ mean squared errors
 - ▶ Seek to control their variation

Reformulation : CUSUM statistic for Gaussian model

$$C_{\tau} = \sqrt{\frac{\tau(n-\tau)}{n}} (\bar{y}_{1:\tau} - \bar{y}_{(\tau+1):n})$$

- ▶ If no changepoint exists at τ , the prefactor ensures the statistic follows a centered normal random variable with variance σ^2 .

Detection of changepoint

$$LR_{\max} = C_{\max}^2 = \max_{\tau \in \{1, \dots, n-1\}} \frac{C_{\tau}^2}{\sigma^2} > \lambda$$

- Declare a change at τ if the squared and variance-normalized CUSUM statistic exceeds a threshold λ .

Computation

- ▶ One critical aspect of modern (multi)-changepoint approaches is their runtime complexity.
- ▶ You want the approach to be computationally scalable.

Stop and Think

Stop and Think

What is the computational complexity of a naive implementation of the CUSUM statistic when computed iteratively along a time series?

An Example

- ▶ Let us compute the CUSUM for the vector:

$$y_{1:4} = (0.8, 1.2, 4.5, 4.3),$$

- ▶ Assuming the observations are Gaussian with $\sigma^2 = 1$.
- ▶ The possible changepoint positions are:

$$\tau \in \{1, 2, 3\}.$$

Empirical Means

- ▶ Before computing the CUSUM statistics for each potential changepoint, we first need to determine the empirical means for each segment.
- ▶ Specifically, for each τ , we compute the means of the subsequences:

$$\bar{y}_{1:\tau} = \frac{1}{\tau} \sum_{i=1}^{\tau} y_i, \quad \bar{y}_{(\tau+1):4} = \frac{1}{4-\tau} \sum_{i=\tau+1}^4 y_i.$$

Computing Means

$$\bar{y}_{1:1} = y_1 = 0.8,$$

$$\bar{y}_{1:2} = \frac{0.8 + 1.2}{2} = 1.0,$$

$$\bar{y}_{1:3} = \frac{0.8 + 1.2 + 4.5}{3} = 2.17,$$

$$\bar{y}_{2:4} = \frac{1.2 + 4.5 + 4.3}{3} = 3.33,$$

$$\bar{y}_{3:4} = \frac{4.5 + 4.3}{2} = 4.4,$$

$$\bar{y}_{4:4} = y_4 = 4.3.$$

CUSUM Statistic

- ▶ The squared CUSUM statistic for each τ is then given by:

$$C_{\tau}^2 = \frac{\tau(4 - \tau)}{4} (\bar{y}_{1:\tau} - \bar{y}_{(\tau+1):4})^2.$$

Computing CUSUM Values

$$C_1^2 = \frac{3 \times 1}{4}(0.8 - 3.33)^2 = 0.75 \times 6.4 = 4.8,$$

$$C_2^2 = \frac{2 \times 2}{4}(1.0 - 4.4)^2 = 11.56,$$

$$C_3^2 = \frac{1 \times 3}{4}(2.17 - 4.3)^2 = 0.75 \times 4.53 = 3.4.$$

Maximum CUSUM Statistic

- ▶ Thus, the maximum CUSUM statistic is:

$$C_{\max}^2 = 11.56 \text{ at } \tau = 2.$$

- ▶ Recall, that to call a changepoint, we would compare $C_{\max}^2$ to a threshold value λ .
- ▶ If $C_{\max}^2 > \lambda$, we conclude that there is a change

Naive Implementation

- ▶ A naive implementation of the CUSUM statistic computes the means $\bar{y}_{1:\tau}$ and $\bar{y}_{\tau+1:n}$ for each τ in $\mathcal{O}(n)$.
- ▶ This leads to an overall computational complexity of $\mathcal{O}(n^2)$.

Efficient Implementation

- ▶ We can be much faster by sequentially computing partial sums:

$$S_t = \sum_{t=1}^n y_t.$$

- ▶ By computing partial sums incrementally, we avoid redundant calculations.
- ▶ This achieves a linear time complexity of $\mathcal{O}(n)$, making the method scalable for large datasets.

Incremental Update of the Mean

- ▶ This incremental update of the mean is key to the efficiency of many algorithms and methods for multiple changepoint detection.
- ▶ Particularly in approaches based on dynamic programming [Auger and Lawrence, 1989, Killick et al., 2012, Rigai, 2015] and approaches based on local optimization or isolation [Fryzlewicz, 2014, Fryzlewicz, 2020, Kovács et al., 2023, Anastasiou and Fryzlewicz, 2022]

Controlling the CUSUM Statistics

We aim to find a threshold such that we declare a changepoint if

$$\max_{\tau < n} C_\tau > \beta$$

- ▶ The choice of the threshold β is crucial.
- ▶ It should be:
 - ▶ large enough to avoid false positives;
 - ▶ small enough to detect true changes.
- ▶ A simple choice is to take β proportional to $\sqrt{\sigma^2 \log(n)}$.
- ▶ Sharper choices are discussed in [Verzelen et al., 2023].

* A Bonferroni-like Bound

Assuming the y_t are all Gaussian with $\sigma^2 = 1$ (without loss of generality), the C_τ are $\mathcal{N}(0, 1)$ and we can use a sub-Gaussian concentration bound.

- ▶ Sub-Gaussian concentration inequality

$$P(|C_\tau| \geq \beta) \leq 2e^{-\frac{\beta^2}{2}}$$

- ▶ Union bound

$$P\left(\max_{\tau < n} |C_\tau| \geq \beta\right) = P\left(\bigcup_{\tau < n} |C_\tau| \geq \beta\right) \leq \sum_{\tau < n} P(|C_\tau| \geq \beta)$$

* Threshold Calculation

- ▶ Union bound (continued)

$$P\left(\max_{\tau < n} |C_\tau| \geq \beta\right) \leq (n-1) P(|C_\tau| \geq \beta) = 2(n-1) e^{-\frac{\beta^2}{2}}$$

- ▶ Taking $\beta = \sqrt{2(\log(n) + \varepsilon)}$

$$P\left(\max_{\tau < n} |C_\tau| \geq \beta\right) \leq 2 \frac{n-1}{n} e^{-\varepsilon} \leq 2 e^{-\varepsilon}$$

- ▶ Control achieved for ε large enough
 - ▶ example: $\varepsilon = 4.6$ yields a probability below 2%

* Detection power

Consider a change of size δ at τ^* :

$$C_{\tau^*} = \sqrt{\frac{\tau^*(n - \tau^*)}{n}} (\delta + \bar{\varepsilon}_{\tau^*+1:n} - \bar{\varepsilon}_{1:\tau^*})$$

- We now aim to estimate the power we get from our crude union-bound threshold

* CUSUM statistic for the true changepoint C_{τ^*}

We consider the data

$$\begin{aligned}y_t &= 0 + \varepsilon_t && \text{for } t \leq \tau^* \\y_t &= \delta + \varepsilon_t && \text{for } t > \tau^* \\&\text{with } \varepsilon_t \sim \mathcal{N}(0, \sigma^2)\end{aligned}$$

Rewriting C_{τ^*} we get:

$$C_{\tau^*} = \delta \sqrt{\frac{\tau^*(n - \tau^*)}{n}} + \sqrt{\frac{\tau^*(n - \tau^*)}{n}} (\bar{\varepsilon}_{\tau^*:n} - \bar{\varepsilon}_{1:\tau^*})$$

* Using Sub-Gaussian Concentration Again

- ▶ With probability at least $1 - 2 \exp(-0.5x^2)$ we have:

$$\left| \sqrt{\frac{\tau^*(n - \tau^*)}{n}} (\bar{\varepsilon}_{\tau^*:n} - \bar{\varepsilon}_{1:\tau^*}) \right| \leq x$$

* Condition for Change Detection

Considering $x = \sqrt{-2 \log(\alpha)}$, we get that if

$$\delta \sqrt{\frac{\tau^*(n - \tau^*)}{n}} > \sqrt{2 \log((n - 1)/\alpha)} + \sqrt{-2 \log(\alpha)}$$

then with probability at least $1 - \alpha$, C_{τ^*} would pass the threshold and we would detect a change.

Key Insights

From this, we get two important pieces of information:

1. The power depends on the position of the change, with changes close to the border being harder to detect.
2. Consider τ to be proportional to n , $\tau = a \cdot n$ (with $a \in (0, 1)$), we detect the change if

$$\delta \sqrt{a(1-a)n} \geq \sqrt{2 \log((n-1)/\alpha)} + \sqrt{-2 \log(\alpha)}.$$

- ▶ this will always happen eventually as $\sqrt{n}$ grows faster than $\log(n)$.

CUSUM simple R code

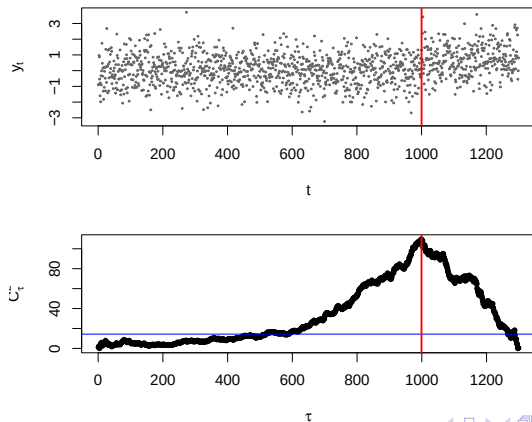
```
# DATA SIMULATION
n1      <- 1000; n2 <- 300
delta   <- 0.6
y        <- c(rnorm(n1), rnorm(n2) + delta)
n        <- n1 + n2
true_cp  <- n1

# CUSUM
S        <- sum(y)
csy      <- cumsum(y)[-n]
rcsy     <- (S - cumsum(y))[-n]

cusum.stat <- csy^2 / (1:(n - 1)) +
              rcsy^2 / ((n - 1):1) -
              S^2 / n
est_cp    <- which.max(cusum.stat)
```

CUSUM simple R plot

```
plot(cusum.stat, pch = 20, xlab = expression(tau),  
     ylab = expression(C[tau]^2))  
abline(h = 2 * log(n), col = "blue")  
abline(v = est_cp, col = "red", lwd = 2)
```



Univariate change-in-mean model : multiple changepoints

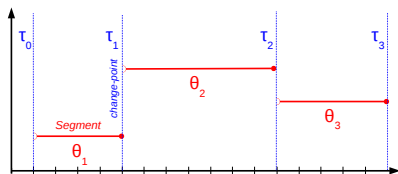
Data $Y_1, \dots, Y_n$

Changepoints $\tau^* = (\tau_1^*, \dots, \tau_D^*)$

Segments $s_d = (\tau_{d-1}^*, \tau_d^*]$

Parameters $\theta = (\theta_1, \dots, \theta_D)$

Model $Y_i \sim \mathcal{F}(\theta_d)$ i.i.d



► Gaussian model : $Y_i \sim \mathcal{N}(\mu_d, \sigma^2)$

$$\sum_{d=1}^{|\tau|} \left(\sum_{\tau_{d-1}+1}^{\tau_d} (Y_i - \hat{\mu}_d)^2 + \lambda \right)$$

Stop and Think

Stop and Think

- ▶ How many possible segmentations are there?
- ▶ From an algorithmic perspective, how can we optimize this?

Binary Segmentation

Heuristic

- ▶ Iterative application of the CUSUM statistic
- ▶ Computational complexity?
- ▶ Good statistical and practical properties
[Fryzlewicz, 2014, Verzelen et al., 2023]

Dynamic programming on changepoints

Exact algorithm

- ▶ A sum over segments
- ▶ Segment cost

$$C_{\tau:n} = \sum_{\tau+1}^n (Y_i - \bar{Y}_{\tau+1:n})^2$$

- ▶ Iteration over changepoints

$$Cost_n = \min_{\tau < n} \{ Cost_{\tau} + C_{\tau:n} + \lambda \}$$

- ▶ Computational complexity?

Dynamic programming on changepoints

Exact algorithm

- ▶ A sum over segments
- ▶ Segment cost

$$C_{\tau:n} = \sum_{i=\tau+1}^n (Y_i - \bar{Y}_{\tau+1:n})^2$$

- ▶ Iteration over changepoints

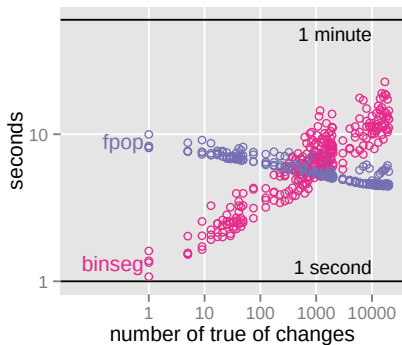
$$Cost_n = \min_{\tau < n} \{ Cost_{\tau} + C_{\tau:n} + \lambda \}$$

- ▶ Computational complexity?
- ▶ Possible pruning: PELT [Killick et al., 2012]
 - ▶ $\mathcal{O}(n)$ if the number of changepoints is proportional to n
 - ▶ $\mathcal{O}(n^2)$ if the number of changepoints is sparse

Dynamic programming on the mean: FPOP [Maidstone et al., 2017]

Exact algorithm

- ▶ FPOP has expected $\mathcal{O}(n \log(n))$ complexity, even in the absence of changepoints
- ▶ Runtime for $n = 10^7$



Statistics: Choosing a penalty

- ▶ Simple: $2\sigma^2|\tau| \log(n)$ [Yao and Au, 1989]
- ▶ Slightly more sophisticated: $2\sigma^2|\tau| (c_1 \log(n/|\tau|) + c_2)$ [Lebarbier, 2005]
- ▶ And many others:
[Zhang and Siegmund, 2007, Verzelen et al., 2023]

Hands-on session

Who's that RNase?

Using changepoint detection to investigate chloroplast RNA processing in *Arabidopsis thaliana*

Let's get started!



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